

Gemini Chat

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [121, 141, 154, 161, 143, 150, 128, 128, 126, 150]

Tap amplitudes (pixels): [176.3887751530692, 180.31361568112376, 194.6175737183053, 199.6121238802894, 185.0, 196.7968495682794, 181.6865432551349, 195.13328778042973, 182.72657168567466, 169.07099100673656, 206.4097865896867]



Based on the extracted features provided, here is the analysis of the subject's performance on the MDS-UPDRS Item 3.4 Finger Tapping test.

Analysis of Extracted Features

1. Speed and Rhythm (Tap Periods)

- **Data:** [121, 141, 154, 161, 143, 150, 128, 128, 126, 150]
- **Observation:** The subject displays noticeable variance in tapping speed. The periods initially increase (slowing down) from 121 to 161 frames over the first four intervals, suggesting a hesitation or slight loss of rhythm. The subject then speeds up (periods dropping to ~126-128) before slowing again at the end.
- **Conclusion:** While there are no complete freezes or halts, the regular rhythm is not maintained consistently. The fluctuation suggests slight hesitations.

2. Amplitude (Tap Amplitudes)

- **Data:** [176.4, 180.3, 194.6, 199.6, 185.0, 196.8, 181.7, 195.1, 182.7, 169.1, 206.4]
- **Observation:** The amplitude is well-maintained throughout the sequence. The taps do not show a

consistent decrement. In fact, the final tap (206.4) is the largest in the sequence. There is no evidence of fatigue or "decrementing amplitude" (a key marker for Parkinsonian bradykinesia).

- **Conclusion:** Amplitude performance is normal.

3. Interruptions/Halts

- **Observation:** There are no extreme outliers in the period data that would indicate a "freeze" or "halt" (which would typically result in a period significantly longer than the others, e.g., 2-3x the average). The variance observed fits the definition of "interruptions or hesitations" rather than "arrests."

Scoring Justification

- **Criteria for 0 (Normal):** Requires "No problems." The rhythm variance here is significant enough to rule out a perfectly normal score.
- **Criteria for 1 (Slight):** This score is applicable if "The regular rhythm is broken with one or two interruptions or hesitations." The period data shows a distinct fluctuation (slowing down then speeding up), which aligns with this criterion.
- **Criteria for 2 (Mild):** Requires "3 to 5 interruptions" or "Mild slowing" or "Amplitude decrements midway." The amplitude does not decrement, and the interruptions are not frequent enough to classify as mild.

Determination: The subject maintains amplitude well but exhibits slight irregularities in rhythm (hesitations/speed variance). This matches the criteria for a "Slight" impairment.

